Suzuki polyhedra in field coordinates

Human Compression project

Abstract

An ordered string marking of a Suzuki group can be normalized to two field coordinates. We identify its generated group by the joint field of definition and prove the intersection property directly. The common orbit theorem then gives both counting formulas of Kiefer and Leemans, including their equality in every allowed field, the prime-degree specialization and the absence of self-dual examples. The finite diagnostics are separate from the uniform written argument.

The source is A. Kiefer and D. Leemans, *On the number of abstract regular polytopes whose automorphism group is a Suzuki simple group Sz(q)*, Journal of Combinatorial Theory, Series A **117** (2010), 1248–1257. The exact consulted author manuscript is dated 27 September 2011 and has twelve pages. The statements below preserve that edition's two principal counting theorems and their stated geometric consequences.

The normalization and generation arguments retain the classical Suzuki-group structure used in that source, including the work of Suzuki and Lüneburg. The earlier rank result of Leemans (2006) is credited at its restatement in the source; it is not an additional source recovered by this application. The shared counting principles are the classical orbit–stabilizer identity, fixed-point counting and Möbius inversion, stated with complete proofs in [UD]. The contribution here is the explicit route from those principles to the normalized data, with the local hypotheses discharged.

1 Retained structure and normalization

Write $q = 2^N$, where $N > 1$ is odd, and $S_N = \mathrm{Sz}(q)$. Use its usual ovoid action, with distinguished points $\infty, 0$, and root group U_q at infinity. Write its central involutions as $z(b)$, $b \in \mathbb{F}_q$, with $z(b)z(c) = z(b + c)$. A Weyl involution w swaps infinity and zero and is defined over $\mathbb{F}_2$.

We retain the following classical Suzuki-group facts. The ovoid and root-group descriptions are the finite specialization of Lüneburg, *Translation planes* (1980), as cited in source Section 2. The maximal-subgroup classification is Suzuki, *On a class of doubly transitive groups* (1962), cited explicitly in source Section 3. The automorphism and centralizer facts belong to the same standard structure theory; their use is retained, not a new classification here:

1. The ovoid action is doubly transitive. Every involution fixes exactly one ovoid point. U_q acts regularly on the points other than infinity.
2. $C_{S_N}(z(b)) = U_q$ for $b \neq 0$, and all involutions in U_q are its central elements $z(b)$. All involutions of S_N are conjugate.
3. The ordered two-point stabilizer is the split torus H_q. Its conjugation acts as $z(b) \mapsto z(tb)$ and $wz(c)w \mapsto wz(t^{-1}c)w$, as t ranges bijectively over $\mathbb{F}_q^*$. This uses a reparametrization of the usual diagonal torus. Its central-coordinate exponent is coprime to $q - 1$.
4. $\mathrm{Aut}(S_N) = S_N \rtimes \langle F \rangle$, where $F : x \mapsto x^2$ has order N. Canonical subfield groups S_d for $d \mid N$ have the corresponding subfield coordinates and ovoids.
5. Proper maximal subgroups are point stabilizers, odd-torus normalizers with cyclic Sylow 2-subgroups, and conjugates of proper subfield groups. A nonidentity odd-order element has a unique maximal odd cyclic torus as its centralizer. The normalizer of such a torus is proper.

6. S_N is nonabelian simple. String C-groups correspond to regular abstract polytopes, with reversing the generator order corresponding to duality.

For clarity, the complete list of possible maximal-subgroup types used here is the source's Table 1. Put $r = \sqrt{2q}$, $\alpha = q + r + 1$ and $\beta = q - r + 1$. Then $|S_N| = q^2(q^2+1)(q-1)$ and

maximal subgroup type	order
$U_q \rtimes C_{q-1}$	$q^2(q-1)$
$D_{2(q-1)}$	$2(q-1)$
$C_\alpha \rtimes C_4$	4α
$C_\beta \rtimes C_4$	4β
$\mathrm{Sz}(2^d)$, N/d prime	$2^{2d}(2^{2d}+1)(2^d-1)$.

Here d is a proper divisor of N, and each index is $|S_N|$ divided by the displayed order. The group U_q has order q^2 and is the source's nonsplit extension of two elementary abelian groups of order q. Only this retained list and the stated involution and torus properties enter the generation and intersection arguments.

The root/torus assertions can equally be read in the standard four-by-four matrix model. Restricting them to a subfield is compatible with the Suzuki field automorphism: since N/d is odd, its exponent restricts to the canonical exponent for $\mathbb{F}_{2^d}$. In particular, the central torus scaling parameter lies in a subfield exactly when its diagonal parameter does. The torus acts faithfully on the nonzero central elements.

Normalization theorem. Inner-conjugacy classes of ordered triples (r_0, r_1, r_2) of involutions with $r_0 \neq r_2$, $[r_0, r_2] = 1$, and $[r_0, r_1] \neq 1$ are in bijection with

$$X_N = (\mathbb{F}_q \setminus \mathbb{F}_2) \times \mathbb{F}_q^*.$$

The representative for (b, c) is

$$\big(z(1),\ wz(c)w,\ z(b)\big).$$

Proof. Conjugate r_0 to $z(1)$. Its commuting involutions are exactly $z(b)$, and excluding the identity and r_0 means $b \neq 0, 1$. Its centralizer U_q fixes every such $z(b)$. The involution r_1 fixes a point other than infinity, since otherwise it would belong to U_q and commute with r_0. There is a unique element of U_q that moves that point to zero. The resulting r_1 is a central involution of the root group at zero, hence $wz(c)w$ with $c \neq 0$. No further element of U_q fixes zero, proving uniqueness. Conversely every displayed pair gives a triple with the stated properties. □

Thus field automorphisms act on normalized coordinates simultaneously:

$$F(b, c) = (b^2, c^2).$$

Reversing the endpoints gives $(z(b), wz(c)w, z(1))$. Normalize its first entry by split-torus scaling $t = b^{-1}$ to obtain the duality action

$$D(b, c) = (b^{-1}, bc).$$

It is an involution commuting with F.

2 Generation and the field of definition

Subfield-position lemma. If a conjugate of a canonical S_d, $d \mid N$, contains $z(1)$ and $wz(c)w$, then it is the canonical S_d itself.

Proof. The embedded ovoid of that conjugate contains infinity and zero, because those are the unique fixed ovoid points of the two involutions. The canonical S_d is doubly transitive on its own ovoid. Changing the conjugator on its S_d side therefore makes it fix the ordered pair $(\infty, 0)$, so it belongs to H_q. Suppose its central scaling is t. The conjugate root center is then $z(t\mathbb{F}_{2^d})$. Since it contains $z(1)$, $t \in \mathbb{F}_{2^d}^*$. The torus/subfield compatibility in Section 1 implies that the conjugator is in $H_{2^d} \leq S_d$. It consequently normalizes S_d, proving the claim. □

Generation theorem. The group generated by the normalized triple is exactly S_d, where

$$\mathbb{F}_{2^d} = \mathbb{F}_2(b, c).$$

Proof. The displayed generators belong to S_d, and $d > 1$ because $b \notin \mathbb{F}_2$. If they generated a proper subgroup of S_d, place it in a maximal subgroup. A point stabilizer cannot contain the triple: its involutions would all fix that point, whereas the first and middle generators fix different points. An odd-torus normalizer cannot contain the four-group $\langle z(1), z(b)\rangle$, because its Sylow 2-subgroups are cyclic. The remaining possibility is a conjugate of S_e for a proper divisor e of d. The position lemma makes it canonical, forcing $b, c \in \mathbb{F}_{2^e}$, a contradiction. □

This proves generation from field coordinates without retaining the source's enumeration formula or its generation-count lemma.

3 The intersection property and rank

The first and middle involutions in a normalized triple do not commute; neither do the middle and last, by their different fixed points. The dihedral group generated by either adjacent pair has odd rotation order. Indeed, if that order had 2-part at least four, its dihedral Sylow 2-subgroup would have non-commuting involutions, impossible inside a Suzuki root group. If the rotation order had 2-part exactly two, its central involution would centralize both generators, placing both in one root group, whose involutions commute. Rotation order two is also excluded by noncommutation.

Suppose these two odd-rotation dihedral subgroups intersect beyond their common middle involution. Their intersection then contains a nontrivial odd-order rotation x. Both rotation subgroups lie in the unique maximal torus $C_{S_d}(x)$. Each adjacent dihedral subgroup normalizes that torus, so all three generators lie in its proper normalizer. This contradicts the generation theorem. Their intersection is consequently exactly the order-two group generated by the middle involution.

The outer four-group intersects either odd-rotation dihedral subgroup in exactly the appropriate endpoint subgroup, by order divisibility. Together these statements give every rank-three subgroup-intersection identity. Every normalized triple is therefore a nondegenerate string C-group for its generated subfield group. Its adjacent product orders are at least three.

There is no string C-group representation of S_N of rank at least four. In such a representation, r_0 commutes with every r_j for $j \geq 2$. These involutions lie in the center of one root group and therefore commute with each other. In particular r_3 commutes with all r_j for $j \geq 2$, as well as with r_0, r_1 by the string relations. This would be a nontrivial central element of S_N. Ranks one and two cannot generate a nonabelian simple group.

Rank three always occurs: take b of full field degree N and $c = 1$. Finally, every involution of a group $S_N \leq G \leq \mathrm{Aut}(S_N)$ maps trivially to G/S_N, since that quotient has odd order. A proper extension cannot be generated by involutions. This recovers the full rank/existence statement of L085 quoted as L074, Theorem 3, without using that theorem as an input.

4 The two counting formulas

Apply Theorem UD.ExactSupportOrbits to the divisor poset of N. The normalization and generation theorems prove that the support is the joint field degree. If $A(d)$ counts data defined over $\mathbb{F}_{2^d}$, the

exact-degree count is

$$E(N) = \sum_{d|N} \mu(N/d)A(d). \tag{E3}$$

Every exact-degree datum has a Frobenius orbit of length N. The additional duality quotient requires the freeness argument that follows; it is not an automatic division by two.

For pairs of distinct commuting involutions, normalized data are just $b \in \mathbb{F}_q \setminus \mathbb{F}_2$. Put

$$\lambda(d) = \frac{1}{d} \sum_{e|d} \mu(d/e)2^e.$$

For $d > 1$, this is the number of Frobenius orbits of elements of exact degree d; the two degree-one elements have already been excluded. Unordered-pair reversal sends b to b^{-1}. Inversion commutes with Frobenius and has no fixed point on these data: $b^2 = 1$ would give $b = 1$ in characteristic two. The acting Frobenius group has odd order N, so UD.OddOrbits proves that reversal fixes no Frobenius orbit. Therefore the number of unordered-pair classes is

$$\boxed{\frac{1}{2} \sum_{d|N,\ d>1} \lambda(d).}$$

This is exactly L074, Theorem 1.

For polyhedra whose full group is S_N, the marked data are X_N with exact joint field degree N. Here

$$A(d) = (2^d - 2)(2^d - 1).$$

Their Frobenius orbits all have length N. The normalized duality $D(b, c) = (b^{-1}, bc)$ is an involution commuting with Frobenius, as proved in Section 1. It has no fixed point because its first coordinate would force $b = 1$. It preserves the joint field, since each of b, c can be recovered from b^{-1}, bc. Thus UD.OddOrbits applies on the exact-support subset, and the free-action consequence of UD.AdmissibleExtensions gives the number of polyhedra up to isomorphism and duality:

$$\boxed{P(N) = \frac{1}{2N} \sum_{d|N} \mu(N/d)(2^d - 2)(2^d - 1).}$$

There are no self-dual examples. The formula also gives the complete polytope count, by the rank restriction in Section 3. For prime N it reduces to $(q - 2)(q - 1)/(2N)$, the source's Theorem 4. For $N = 3, 5, 9, 15$ it gives respectively $7, 93, 14476, 35788085$.

5 Equivalence with the source's nested formula

This section proves equivalence for every odd N, not just agreement of a finite table. Let $Q(N)$ be the number of Frobenius orbits on all of X_N, including data of smaller joint degree.

Apply UD.ExtensionOrbits to the projection onto the first coordinate, and stratify by its degree $n > 1$. There are $\lambda(n)$ first-coordinate orbits. After choosing a representative, its stabilizer is generated by F^n and acts on $\mathbb{F}_q^*$. For $m \mid N/n$, the number of points fixed by $(F^n)^m$ is $2^{nm} - 1$. Möbius inversion gives

$$\psi(n, N) = \sum_{m|N/n} \frac{1}{m} \sum_{e|m} \mu(m/e)(2^{ne} - 1)$$

orbits of this stabilizer. Therefore

$$Q(N) = \sum_{n|N,\ n>1} \lambda(n)\psi(n,N).$$

Alternatively stratify by the joint field degree d. Orbits on exact-degree-d pairs have length d, and their Frobenius equivalence is the same whether computed in $\mathbb{F}_{2^d}$ or the larger field. Thus $Q(N) = \sum_{d|N} E(d)/d$. Möbius inversion now gives

$$\frac{1}{2}\sum_{d|N} \mu(N/d)Q(d) = \frac{E(N)}{2N} = P(N).$$

The expression on the left is precisely L074, Theorem 2. This derives its full parameter range and nondegeneracy conclusion from the marked field theorem, with all field-automorphism orbit corrections included.

6 Source statements and retained background

The preceding proof also locates the source's auxiliary counting statements. Lemma 7 is the divisor-poset instance of Möbius inversion proved in UD.ExactSupportOrbits; Lemma 8 is the formula for λ; Lemma 9 is the stabilizer-fiber formula for ψ; and Lemma 10 is $Q(N)/2$, before removing data of smaller joint field degree. Remark 1 is the prime-degree specialization. The source's second example follows by substituting $N = 15$: the values of λ at $3, 5, 15$ are $2, 6, 2182$, the corresponding values of $\psi(n, 15)$ are $6559, 10943, 32767$, and $Q(15)/2 = 35788185$. Removing the smaller-degree counts 7 and 93 gives 35788085. Its first example illustrates the same stabilizer $\langle F^n \rangle$: a degree-three coordinate in $\mathbb{F}_{2^{15}}$ is fixed by F^3, whereas a general second coordinate is not.

The earlier structural lemmas have a different status. The source attributes Lemmas 1–3 to Lüneburg (1980), Lemma 4 to Leemans (1998), and Table 1 to Suzuki (1962); those attributions are retained here. Source Lemma 1 follows from the cyclic automorphism group of a finite field: a Frobenius power F^j has square F exactly when $2j \equiv 1 \pmod{N}$, which has a unique solution for odd N and none for even N. The finite ovoid action and root-group consequences of Lemma 2, the ovoid size in Lemma 3, and the maximal-subgroup classification in Table 1 are retained in Section 1. We do not rederive Lemma 2's arbitrary-field Bruhat description or Lemma 3's assertion about plane sections of general ovoids. Those broader statements are not premises of our counting proof. Lemma 4's torus-order coprimality is likewise not used in place of the intersection argument: Section 3 uses uniqueness of the centralizer torus and proves the intersection directly. Lemma 5's involution count is $(q^2+1)(q-1)$ because each of the q^2+1 ovoid points supports $q-1$ central root involutions and every involution fixes exactly one point; its conjugacy assertion is among the retained structure facts. Lemma 6 is the classical Fermat congruence, and the definitions of ovoids, Möbius inversion and the counting function are inherited notation. This inventory distinguishes recovery of the principal classification from reproving every background statement printed in the source.

Verification and references

The companion appendix preserves both complete finite diagnostics: the finite-field enumeration and formula comparisons, and the matrix calculation for all normalized triples at $q = 8$. They do not prove a statement for arbitrary odd field exponent. The written normalization, subfield-position, intersection and counting proofs provide that conclusion.

[UD] *Stabilizers, extensions and the enumeration of finite geometries*, common paper in this topic. Its full text is part of the proof dependencies. The classical inputs used in Section 1 are H. Lüneburg, *Translation planes*, Springer, 1980, and M. Suzuki, *On a class of doubly transitive groups*, Annals of Mathematics **75** (1962), 105–145. The earlier theorem restated as source Theorem 3 is D. Leemans,

Almost simple groups of Suzuki type acting on polytopes, Proceedings of the American Mathematical Society **134** (2006), 3649–3651. The counting result is Kiefer–Leemans; these structural inputs keep their earlier attribution. Fermat's congruence and Möbius inversion are not results newly attributed to Kiefer and Leemans or to this project. The standard string-C-group/polytope correspondence is retained from the established theory of abstract regular polytopes.